CASE STUDY: REDUCING SHIPPING DAMAGE

Packaging System Redesign for Enhanced Logistics Reliability

Executive Summary

Frequent transportation-related damage was causing project disruptions and high replacement costs. By shifting the focus from "handling errors" to **System Reliability**, I led a packaging redesign that replaced failing materials with high-density solutions, significantly reducing transit damage and rework.

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1. CONTEXT & CHALLENGE

The organization was experiencing recurring glass breakage and frame scratches during transit. These failures resulted in costly site reworks and delayed installation schedules.

- **The Problem:** Damage was historically blamed on "poor handling," but a technical review suggested the packaging system itself was under-engineered for real-world transport loads.
- **Systemic Gaps:**
  - Existing low-density Styrofoam spacers were crushing under the weight of the load.
  - Spacer failure allowed internal movement, causing parts to collide.
  - Lack of a validated packaging standard for long-distance logistics.

2. MANAGEMENT & TECHNICAL APPROACH

I approached this as a **Material and System Reliability** issue rather than a behavioral one.

- **Failure Analysis:** Conducted a deep dive into damage patterns to identify the specific point of failure in the packaging sequence.
- **Observation:** Performed "Gemba" at loading and unloading zones to see how the spacers performed under actual stress.
- **Root Cause Identification:** Proved that the material limitation (density of the spacers) was the primary driver of downstream damage.

3. THE SOLUTION: SYSTEM REDESIGN

I led the transition to a more robust packaging architecture:

1. **Material Innovation:** Proposed and validated the use of **High-Density Honeycomb Cardboard** spacers to replace Styrofoam.
2. **Supplier Coordination:** Managed the testing phase with suppliers to ensure the new material met technical load-bearing requirements.
3. **Validation Trials:** Conducted real-world trials to compare protective performance under varied transport conditions.

4. **Standardization:** Established a new "Shipment Packaging Standard" to ensure consistency across all product lines.

4. QUANTIFIED IMPACT

Key Performance Indicator	Baseline (Old System)	New Result (Redesign)
Transit Damage Rate	High / Recurring	**Significantly Reduced**
Replacement Costs	Excessive	**Minimized**
Delivery Quality	Unreliable	**Consistent / High-Standard**
System Reliability	Weak (Material Failure)	**Robust (Engineered Solution)**

5. LEADERSHIP TAKEAWAYS

- **Material Selection is Strategy:** Small design choices in packaging can prevent million-dollar downstream costs.
- **Data Over Blame:** Shifting the narrative from "worker error" to "system design" allows for permanent, technical fixes.
- **Logistics Integrity:** A product is only as good as the condition in which it arrives at the site.

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Expertise: Operational Excellence & Technical Problem Solving